Arul Rhik Mazumder

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Education

Carnegie Mellon University **Aug. 2023 – May 2026 (expected)**

Bachelor of Science in Computer Science, Dean's List High Honors *Pittsburgh, PA*

- GPA: 3.8 / 4.0.
- Relevant Coursework: Probabilistic Graphical Models, Quantum Computing, Deep Learning Systems, Parallel Architecture, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization
- Activities: President and Founder of CMU Quantum, Quant Club, ACM @ CMU, Project Olympus, Fencing Club.

Research/Work Experience

Waterloo Institute for Quantum Computing (IQC) **May 2026 – Aug. 2026**

Undergraduate Researcher *Waterloo, ON, Canada*

- Selected for the Undergraduate School on Experimental Quantum Information Processing (USEQIP) and an Undergraduate Research Award (URA) at IQC. Will research quantum algorithms under Michele Mosca following two weeks of training in experimental quantum information processing. Supported by a MITACS Globalink Scholarship.

Caltech Institute for Quantum Information and Matter (IQIM) **May 2025 – Aug. 2025**

Undergraduate Researcher *Pasadena, CA*

- Conducted research on resource-efficient quantum matrix processing using commutator scaling, achieving significant improvements in algorithmic efficiency. Collaborative work with Samson Wang and James Watson. Accepted to TQC 2026, Presentation and Poster showcased at IQIM; report under revision for paper submission here.

Quantum Technologies Group **May 2023 – Present**

Research Intern *Pittsburgh, PA*

- Created a tutorial on quantum optimization for IBM and D-Wave machines, with a GitHub repo; presented at INFORMS 2025 and published in *Tutorials in Operations Research*.

George Mason University Algorithms Lab **May 2022 – May 2023**

Research Intern *Fairfax, VA*

- Developed a quantum string-matching algorithm in Qiskit, compared to classical methods; published in *ACM AISS '22*.
- Created a hybrid quantum-classical TSP algorithm, outperforming classical approaches; published in *IEEE IPDPS 2023*.
- Developed a quantum algorithm for the Ship Rerouting Problem, tested on real-world networks, outperforming classical solvers; published in *Journal of Student-Scientists' Research, 2024*, with an extension preprinted on arXiv.

Leadership & Outreach in Quantum

CMU Quantum **August 2024 – Present**

Founder and President *Pittsburgh, Pennsylvania*

- Founded and grew a quantum computing club to 200+ members with 10+ leaders, securing sponsorships from General Motors and Boeing. Serve as Head of Leadership, liaising with external organizations, developing educational materials (pdfs/slides), organizing workshops with industry leaders (IBM, IonQ, Quantinuum), and leading hackathon tracks and sponsored trips, including the Quantum Leap Career Nexus.

BlueQubit **October 2025**

Technical Writer and Research Intern *Los Angeles, California*

- Reviewed and synthesized BlueQubit research to produce accessible, technically accurate blog posts for the broader quantum community (e.g., Quantum Advantage You Can Actually Verify: Inside Peaked Circuits on heuristic quantum advantage with peaked circuits).
- Benchmarked and analyzed BlueQubit simulators against external platforms and conducted runtime scaling studies across multiple simulation backends, generating insights that informed research-driven product decisions and enabled the engineering team to improve tooling and simulator cost estimators.

Duke Software-Tailored Architectures for Quantum Codesign (STAQ) **June 2025**

NSF-Funded Invited Scholar *Durham, North Carolina*

- Completed a PhD-level, week-long quantum computing seminar featuring advanced lectures on quantum simulation, error correction, and architecture, taught by leading researchers including Peter Love, Kenneth Brown, and Yu Tong.
- Visited the Duke Quantum Center to learn about experimental research on ion traps and neutral atoms.

Publications and Ongoing Work

Peer-Reviewed Publications

1. **A. R. Mazumder**, J. Watson, S. Wang. ”Resource-Efficient Quantum Matrix Processing with Commutator Scaling.” *TQC*, 2026.
2. **A. R. Mazumder**, S. Tayur. ”Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers.” *Tutorials in Operations Research: Advances in Analytics and Operations Research: Improving Decisions to Secure the Future*, INFORMS, 2025, pp. 145-183. [PDF] from arxiv.org
3. M. Zhao, **A. R. Mazumder**, F. Li. ”Improved Classical and Quantum Algorithms for Shipment Rerouting Problems.” *Journal of Student-Scientists' Research*, 2024.
4. **A. R. Mazumder**, A. Sen, U. Sen. ”Benchmarking Metaheuristic-Integrated QAOA Against Quantum Annealing.” *Lecture Notes in Networks and Systems*, 2024.
5. A. Sen, **A. R. Mazumder**, U. Sen. ”Differential Evolution Algorithm Based Hyperparameter Selection of Transformer Neural Networks.” *IEEE SSCI*, 2023.
6. A. Sen, **A. R. Mazumder**, D. Dutta, U. Sen, P. Syam, S. Dhar. ”Comparative Evaluation of Metaheuristics for Short-Term Weather Forecasting.” *ECTA*, 2023.
7. F. Li, **A. R. Mazumder**. ”An Adaptive Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem.” *IEEE IPDPS*, 2023.
8. M. Gao, R. Huang, **A. R. Mazumder**, F. Li. ”Comparisons of Classic and Quantum String Matching Algorithms.” *ACM AISS*, 2022.

Preprints

1. F. Li, **A. R. Mazumder**, M. Zhao. ”Quantum Annealing Approaches to Shipment Rerouting Problems.” *arXiv preprint arXiv:2501.05624*, 2025.

Awards and Honors

MIT IQuHacks – 2nd Place on NVIDA Track (2026)
Selected among 90+ teams for a scalable variational quantum approach to the low-autocorrelation binary sequence problem. Introduced heuristic improvements for faster convergence to high-quality solutions and used Pauli Correlation Encodings to compress the problem into fewer qubits, enabling variational circuits to solve instances more than twice the size of prior work.

Yale Quantum Institute Grand Prize – 2nd Place Overall, YQuantum Advancement Track Winner (2025)
Awarded for reconstructing density operators from Wigner functions using convex optimization and CNN-based denoising. Presented at the QuantumCT Industry Collaboration Forum, competing among 300 participants from 30+ universities.

Summer Invitational Correlation One & Citadel Securities Datathon - 2nd Place Overall (2024)
Selected as one of the top 150 from 3,000+ participants. Developed a pairs-trading strategy linking the processed food and healthcare sectors, achieving a Sharpe ratio of 1.58. Judged by Citadel Quantitative Analysts and awarded a cash prize.

Best Campus Infrastructure Hack – HackCMU (2023)
Developed ”Eco-Bin,” an AI-driven waste sorting system using computer vision and deep learning (95% accuracy). Recognized by the CMU Graduate Student Association among 200+ participants. Selected to pilot the system with CMU Dining Services.

Regeneron Science Talent Search Scholar: Top 300 Teen Scientists (2023)
Recognized as one of the top 300 scholars in the nation's premier science and math competition for high school seniors, selected from over 2,000 applicants across 712 schools in 46 states, Puerto Rico, Guam, and 10 other countries.

Olympiads: USA(J)MO Qualifier, USAPhO Bronze Medal, USACO Gold, USNCO National Finalist

Technical Skills

Programming: Python, Julia, C++, Java, TensorFlow, PyTorch, CVXPY, PICOS
Quantum Toolkits: IBM Q Experience, D-Wave Ocean SDK, BlueQubit, Qiskit, Pennylane
Systems: Linux, Git, CUDA, MPI, OpenMP, pthreads

References

Available upon request.